STAT-S 670: EXPLORATORY DATA ANALYSIS

FINAL-PROJECT: CUSTOMER SEGMENTATION ANALYSIS

Executive Summary

In the rapidly evolving marketplace, understanding consumer behavior is paramount for businesses aiming to optimize their marketing strategies. This project leverages the Customer Personality Dataset from Kaggle to analyze the spending behaviors of customers based on their income, age, and marital status. By employing advanced statistical and machine learning techniques, we identified distinct patterns and influences that these demographic factors have on spending, which can greatly assist businesses in enhancing their customer engagement and segmentation strategies.

In the preliminary phase of our analysis, we explored the direct relationship between customer income and spending habits. Our findings reveal a pronounced positive correlation—higher income levels generally correspond with increased spending. Further analysis segmented by age groups adds nuance to these observations. Specifically, spending tends to increase with income across all age groups, but patterns vary significantly among different demographics. Notably, younger customers (35 and below) exhibit more conservative spending regardless of income, suggesting that factors other than financial capacity may influence their purchasing decisions. In contrast, customers aged 36-45 exhibits a sharp rise in expenditure as their income grows, reflecting their high responsiveness to increased financial power. Meanwhile, the eldest demographic, those aged 66-75, showed a leveling off in spending beyond a certain income level, likely due to fixed retirement incomes or more conservative spending habits. These insights indicate that while income is a significant determinant of spending, age-related factors also play a crucial role in shaping consumer behavior. Further extending our analysis to include marital status, our findings indicate that this variable does not significantly affect the relationship between income and spending. Regardless of being single, married, divorced, or widowed, all groups exhibited a consistent pattern of increased spending as income rose. This uniformity suggests that marital status, unlike age, might not be a critical factor in predicting spending habits. This finding provides a simpler landscape for marketers targeting based on income levels alone, potentially streamlining demographic considerations in marketing strategies. To further refine our understanding, we conducted an extensive segmentation of the customer base, identifying distinct clusters based on their purchasing behaviors, demographic details, and responsiveness to marketing efforts. This segmentation revealed diverse customer groups each characterized by unique spending behaviors and potential needs. The project utilized Principal Component Analysis (PCA) to reduce the dataset dimensionality and focus on significant variables influencing customer behavior. K-means clustering was then applied to segment the customer base into meaningful groups, providing a clear depiction of distinct purchasing behaviors and preferences. We successfully segmented our customer base into three distinct groups using PCA and K-means clustering:

Emerging Economicals: This segment, primarily younger and lower-income, shows cautious spending behavior with high digital engagement but low transaction values.

Established Balancers: Customers with average income levels, showcasing balanced spending across various product categories. They are characterized by their loyalty and a diverse use of purchasing channels.

High-Income Enthusiasts: Predominantly high earners with a preference for premium products. They spend significantly across all categories, particularly on wines and meats. Despite their high spending, they tend not to utilize deals, indicating a lower sensitivity to price or promotions.

In conclusion, our analysis has provided a comprehensive overview of how demographic factors influence spending behaviors, offering actionable insights for businesses to refine their marketing strategies.